NONLINEAR WAVE THEORY SEMINAR NOTES

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ABSTRACT. These notes collect problems, computations, and numerical experiments from the seminar-style course *Nonlinear Wave Theory* in the Spring semester of the 2025–2026 academic year. Since regular lectures were interrupted, the notes record problems suggested by Professor Liming Ling, including gauge transformations, squared eigenfunctions at Weierstrass points, and numerical reproduction of one-mode anomalous-wave recurrence for the focusing NLS equation.

Some reference papers, arXiv source files, and MATLAB scripts are kept in the repository for study and reproducibility.

HW 1: Two-Step Gauge Transformation **2026-04-22**

Consider the system

$$\mathrm{i}\Psi_t = \frac{1}{2}Q^2\Psi - (\Psi^\dagger\Psi)\Psi + \frac{\Delta}{2}\sigma_3\Psi, \qquad \Psi = (\Psi_1, \Psi_2)^\top, \tag{1.1}$$

$$Q = -\mathrm{i}\partial_x + \alpha A(x), \qquad A(x) = \begin{pmatrix} 0 & \mathrm{e}^{-2\mathrm{i}kx} \\ \mathrm{e}^{2\mathrm{i}kx} & 0 \end{pmatrix}. \tag{1.2}$$

Here

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{1.3}$$

denote the Pauli matrices.

We first compute the square of Q. Since

$$A(x)^2 = \mathbb{I}_2, \qquad A_x(x) = \begin{pmatrix} 0 & -2\mathrm{i}k\,\mathrm{e}^{-2\mathrm{i}kx} \\ 2\mathrm{i}k\,\mathrm{e}^{2\mathrm{i}kx} & 0 \end{pmatrix}, \tag{1.4}$$

we obtain

$$\begin{aligned} Q^2 &= \left(-\mathrm{i}\partial_x + \alpha A\right)^2 \\ &= -\partial_x^2 - \mathrm{i}\alpha A\partial_x - \mathrm{i}\alpha\partial_x(A\,\cdot) + \alpha^2 A^2 \\ &= -\partial_x^2 - \mathrm{i}\alpha A\partial_x - \mathrm{i}\alpha(A_x + A\partial_x) + \alpha^2 A^2 \\ &= -\partial_x^2 - 2\mathrm{i}\alpha A\partial_x - \mathrm{i}\alpha A_x + \alpha^2. \end{aligned} \tag{1.5}$$

Substituting (1.5) into (1.1) yields

$$\mathrm{i}\Psi_t = -\frac{1}{2}\Psi_{xx} - \mathrm{i}\alpha A\Psi_x - \frac{\mathrm{i}}{2}\alpha A_x\Psi + \frac{1}{2}\alpha^2\Psi - |\Psi|^2\Psi + \frac{\Delta}{2}\sigma_3\Psi. \tag{1.6}$$

Define the gauge transformation

$$\Psi(x,t) = \Lambda(x,t)\psi(x,t), \qquad \Lambda(x,t) = \mathrm{e}^{-\frac{\mathrm{i}}{2}(\alpha^2+k^2)t}\mathrm{e}^{-\mathrm{i}kx\sigma_3}. \tag{1.7}$$

Since $\sigma_3^2 = \mathbb{I}_2$, the matrix exponential may be written as

$$\mathrm{e}^{-\mathrm{i}kx\sigma_3} = \begin{pmatrix} \mathrm{e}^{-\mathrm{i}kx} & 0 \\ 0 & \mathrm{e}^{\mathrm{i}kx} \end{pmatrix}, \tag{1.8}$$

and therefore Λ is unitary:

$$\Lambda^\dagger\Lambda = \mathbb{I}_2. \tag{1.9}$$

Date: June 9, 2026.

Consequently,

$$|\Psi|^2 = \Psi^\dagger\Psi = \psi^\dagger\Lambda^\dagger\Lambda\psi = \psi^\dagger\psi = |\psi|^2. \tag{1.10}$$

Differentiating (1.7) gives

$$\Lambda_t = -\frac{\mathrm{i}}{2}(\alpha^2+k^2)\Lambda, \qquad \Lambda_x = -\mathrm{i}k\sigma_3\Lambda. \tag{1.11}$$

Hence

$$\Psi_t = \Lambda_t\psi + \Lambda\psi_t = \Lambda\left(-\frac{\mathrm{i}}{2}(\alpha^2+k^2)\psi + \psi_t\right), \tag{1.12}$$

$$\Psi_x = \Lambda_x\psi + \Lambda\psi_x = \Lambda\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right), \tag{1.13}$$

and

$$\begin{aligned}\Psi_{xx} &= \partial_x[\Lambda\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right)]\\ &= \Lambda_x\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right) + \Lambda\left(-\mathrm{i}k\sigma_3\psi_x + \psi_{xx}\right)\\ &= \Lambda\left[-\mathrm{i}k\sigma_3\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right) - \mathrm{i}k\sigma_3\psi_x + \psi_{xx}\right]\\ &= \Lambda\left(-k^2\psi - 2\mathrm{i}k\sigma_3\psi_x + \psi_{xx}\right).\end{aligned} \tag{1.14}$$

Next we compute the conjugation of $A(x)$ by Λ. Since the scalar phase $\mathrm{e}^{-\frac{\mathrm{i}}{2}(\alpha^2+k^2)t}$ commutes with every matrix, we have

$$\begin{aligned}\Lambda^{-1}A(x)\Lambda &= \mathrm{e}^{\mathrm{i}kx\sigma_3}A(x)\mathrm{e}^{-\mathrm{i}kx\sigma_3}\\ &= \begin{pmatrix}\mathrm{e}^{\mathrm{i}kx} & 0\\ 0 & \mathrm{e}^{-\mathrm{i}kx}\end{pmatrix}\begin{pmatrix}0 & \mathrm{e}^{-2\mathrm{i}kx}\\ \mathrm{e}^{2\mathrm{i}kx} & 0\end{pmatrix}\begin{pmatrix}\mathrm{e}^{-\mathrm{i}kx} & 0\\ 0 & \mathrm{e}^{\mathrm{i}kx}\end{pmatrix}\\ &= \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \sigma_1.\end{aligned} \tag{1.15}$$

Similarly,

$$\Lambda^{-1}A_x(x)\Lambda = \begin{pmatrix}0 & -2\mathrm{i}k\\ 2\mathrm{i}k & 0\end{pmatrix} = 2k\begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix} = 2k\sigma_2 = 2\mathrm{i}k\sigma_1\sigma_3. \tag{1.16}$$

Moreover,

$$\Lambda^{-1}\sigma_3\Lambda = \sigma_3, \tag{1.17}$$

since σ_3 commutes with $\mathrm{e}^{-\mathrm{i}kx\sigma_3}$.

Substituting (1.12), (1.13), and (1.14) into (1.6), we obtain

$$\begin{aligned}&\mathrm{i}\Lambda\left(-\frac{\mathrm{i}}{2}(\alpha^2+k^2)\psi + \psi_t\right)\\ &= -\frac{1}{2}\Lambda\left(-k^2\psi - 2\mathrm{i}k\sigma_3\psi_x + \psi_{xx}\right)\\ &\quad - \mathrm{i}\alpha A\Lambda\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right) - \frac{\mathrm{i}}{2}\alpha A_x\Lambda\psi + \frac{1}{2}\alpha^2\Lambda\psi\\ &\quad - |\psi|^2\Lambda\psi + \frac{\Delta}{2}\sigma_3\Lambda\psi.\end{aligned} \tag{1.18}$$

Multiplying (1.18) from the left by Λ^{-1} and using (1.15)–(1.17) gives

$$\begin{aligned}\frac{1}{2}(\alpha^2+k^2)\psi + \mathrm{i}\psi_t &= \frac{1}{2}k^2\psi + \mathrm{i}k\sigma_3\psi_x - \frac{1}{2}\psi_{xx}\\ &\quad - \mathrm{i}\alpha\sigma_1\left(-\mathrm{i}k\sigma_3\psi + \psi_x\right) - \mathrm{i}\alpha k\sigma_2\psi\\ &\quad + \frac{1}{2}\alpha^2\psi - |\psi|^2\psi + \frac{\Delta}{2}\sigma_3\psi.\end{aligned} \tag{1.19}$$

Using the identity

$$\sigma_1\sigma_3 = -\mathrm{i}\sigma_2, \tag{1.20}$$

we find

$$-\mathrm{i}\alpha\sigma_1(-\mathrm{i}k\sigma_3\psi) = \mathrm{i}\alpha k\sigma_2\psi, \tag{1.21}$$

which exactly cancels the term $-\mathrm{i}\alpha k\sigma_2\psi$ in (1.19). After cancellation of the constant terms, we arrive at

$$\mathrm{i}\psi_t = -\frac{1}{2}\psi_{xx} + \mathrm{i}(k\sigma_3 - \alpha\sigma_1)\psi_x - |\psi|^2\psi + \frac{\Delta}{2}\sigma_3\psi. \tag{1.22}$$

We now introduce the second gauge transformation

$$\psi(x,t) = V(x,t)q(x,t), \qquad V(x,t) = \mathrm{e}^{\frac{\mathrm{i}}{2}(\alpha^2+k^2)t}\, B\, \mathrm{e}^{-\mathrm{i}\kappa x\sigma_3}, \qquad \kappa = \sqrt{\alpha^2+k^2}, \tag{1.23}$$

where

$$B = \begin{pmatrix} -\sin\varphi & -\cos\varphi \\ \cos\varphi & -\sin\varphi \end{pmatrix} = -\sin\varphi\,\mathbb{I}_2 + \cos\varphi\,\sigma_1\sigma_3. \tag{1.24}$$

The angle φ is chosen so that

$$2\varphi = \arctan\frac{k}{\alpha} - \frac{\pi}{2}. \tag{1.25}$$

Hence

$$\tan(2\varphi) = -\frac{\alpha}{k}, \tag{1.26}$$

and therefore

$$\sin(2\varphi) = -\frac{\alpha}{\sqrt{\alpha^2+k^2}}, \qquad \cos(2\varphi) = \frac{k}{\sqrt{\alpha^2+k^2}}. \tag{1.27}$$

Since B is independent of x and t, differentiating (1.23) yields

$$\psi_t = V\left(\frac{\mathrm{i}}{2}\kappa^2 q + q_t\right), \tag{1.28}$$

$$\psi_x = V\left(-\mathrm{i}\kappa\sigma_3 q + q_x\right), \tag{1.29}$$

$$\psi_{xx} = V\left(-\kappa^2 q - 2\mathrm{i}\kappa\sigma_3 q_x + q_{xx}\right). \tag{1.30}$$

Next we compute several algebraic identities associated with B. Since

$$B^\top = \begin{pmatrix} -\sin\varphi & \cos\varphi \\ -\cos\varphi & -\sin\varphi \end{pmatrix}, \tag{1.31}$$

a direct computation gives

$$B^\top B = \mathbb{I}_2. \tag{1.32}$$

Consequently,

$$V^\dagger V = \mathbb{I}_2, \tag{1.33}$$

and hence the nonlinear term is preserved:

$$|\psi|^2 = \psi^\dagger\psi = q^\dagger V^\dagger V q = q^\dagger q = |q|^2. \tag{1.34}$$

We now compute the conjugation of σ_3 by B. Using (1.27), one finds

$$B^\top\sigma_3 B = \begin{pmatrix} -\cos(2\varphi) & \sin(2\varphi) \\ \sin(2\varphi) & \cos(2\varphi) \end{pmatrix} = -\frac{1}{\sqrt{\alpha^2+k^2}}\begin{pmatrix} k & \alpha \\ \alpha & -k \end{pmatrix}. \tag{1.35}$$

Using (1.24), we compute

$$\begin{aligned} B^\top(k\sigma_3 - \alpha\sigma_1)B &= kB^\top\sigma_3 B - \alpha B^\top\sigma_1 B \\ &= \begin{pmatrix} -k\cos(2\varphi) + \alpha\sin(2\varphi) & k\sin(2\varphi) + \alpha\cos(2\varphi) \\ k\sin(2\varphi) + \alpha\cos(2\varphi) & k\cos(2\varphi) - \alpha\sin(2\varphi) \end{pmatrix}. \end{aligned} \tag{1.36}$$

Substituting (1.27) gives

$$B^\top(k\sigma_3-\alpha\sigma_1)B=\begin{pmatrix}-\kappa & 0\\ 0 & \kappa\end{pmatrix}=-\kappa\sigma_3. \tag{1.37}$$

Substituting (1.28)–(1.30) into (1.22), we obtain

$$\begin{aligned}\mathrm{i}V\left(\frac{\mathrm{i}}{2}\kappa^2q+q_t\right)&=-\frac{1}{2}V\left(-\kappa^2q-2\mathrm{i}\kappa\sigma_3q_x+q_{xx}\right)\\&\quad+\mathrm{i}(k\sigma_3-\alpha\sigma_1)V\left(-\mathrm{i}\kappa\sigma_3q+q_x\right)-|q|^2Vq+\frac{\Delta}{2}\sigma_3Vq.\end{aligned} \tag{1.38}$$

Multiplying (1.38) from the left by V^{-1} yields

$$\begin{aligned}-\frac{1}{2}\kappa^2q+\mathrm{i}q_t=&\frac{1}{2}\kappa^2q+\mathrm{i}\kappa\sigma_3q_x-\frac{1}{2}q_{xx}\\&+\mathrm{i}V^{-1}(k\sigma_3-\alpha\sigma_1)V\left(-\mathrm{i}\kappa\sigma_3q+q_x\right)\\&-|q|^2q+\frac{\Delta}{2}V^{-1}\sigma_3Vq.\end{aligned} \tag{1.39}$$

Since

$$V^{-1}(k\sigma_3-\alpha\sigma_1)V=\mathrm{e}^{\mathrm{i}\kappa x\sigma_3}B^\top(k\sigma_3-\alpha\sigma_1)B\mathrm{e}^{-\mathrm{i}\kappa x\sigma_3}, \tag{1.40}$$

equation (1.37) implies

$$V^{-1}(k\sigma_3-\alpha\sigma_1)V=-\kappa\sigma_3. \tag{1.41}$$

Substituting (1.41) into (1.39), the first-order derivative terms cancel:

$$\mathrm{i}q_t=-\frac{1}{2}q_{xx}-|q|^2q+\frac{\Delta}{2}V^{-1}\sigma_3Vq. \tag{1.42}$$

Finally,

$$\begin{aligned}V^{-1}\sigma_3V&=\mathrm{e}^{\mathrm{i}\kappa x\sigma_3}B^\top\sigma_3B\mathrm{e}^{-\mathrm{i}\kappa x\sigma_3}\\&=\mathrm{e}^{\mathrm{i}\kappa x\sigma_3}\frac{-1}{\sqrt{\alpha^2+k^2}}\begin{pmatrix}k & \alpha\\ \alpha & -k\end{pmatrix}\mathrm{e}^{-\mathrm{i}\kappa x\sigma_3}.\end{aligned} \tag{1.43}$$

Therefore,

$$\mathrm{i}q_t=-\frac{1}{2}q_{xx}-|q|^2q+\frac{\Delta}{2}\mathrm{e}^{\mathrm{i}\kappa x\sigma_3}\frac{-1}{\sqrt{\alpha^2+k^2}}\begin{pmatrix}k & \alpha\\ \alpha & -k\end{pmatrix}\mathrm{e}^{-\mathrm{i}\kappa x\sigma_3}q. \tag{1.44}$$

HW 2: The squared eigenfunctions on Weierstrass points **2026-04-29**

Let $B\in\mathbb{C}^{g\times g}$ be a symmetric matrix with positive definite imaginary part. The associated Riemann theta function is defined by

$$\Theta(\boldsymbol{z},B)=\sum_{\boldsymbol{n}\in\mathbb{Z}^g}\exp\Big(\pi\mathrm{i}\,\boldsymbol{n}^\top B\boldsymbol{n}+2\pi\mathrm{i}\,\boldsymbol{n}^\top\boldsymbol{z}\Big),\qquad \boldsymbol{z}\in\mathbb{C}^g. \tag{2.1}$$

For $k=1,\dots,g$, let

$$\boldsymbol{e}_k=(0,\dots,0,1,0,\dots,0)^\top$$

denote the k-th standard basis vector of $\mathbb{R}^g$. The theta function satisfies the quasi-periodicity relations

$$\Theta(\boldsymbol{z}+\boldsymbol{e}_k,B)=\Theta(\boldsymbol{z},B), \tag{2.2}$$

and

$$\Theta(\boldsymbol{z}+B\boldsymbol{e}_k,B)=\exp(-2\pi\mathrm{i}\,z_k-\pi\mathrm{i}\,B_{kk})\,\Theta(\boldsymbol{z},B), \tag{2.3}$$

where z_k denotes the k-th component of $\boldsymbol{z}$.

Consider the genus-two hyperelliptic curve

$$\mathscr{F}_2 = \left\{ P = (\lambda, y) \;\middle|\; y^2 = \prod_{j=1}^{3} (\lambda - \lambda_j)(\lambda - \lambda_j^*) \right\}, \tag{2.4}$$

with $\lambda \in \mathbb{CP}^1$.

In Case 1, we assume that

$$\lambda_1, \lambda_2, \lambda_3 \in \mathrm{i}\mathbb{R}, \qquad \Im\lambda_3 > \Im\lambda_2 > \Im\lambda_1 > 0.$$

Let $\{a_1, a_2, b_1, b_2\}$ be a canonical homology basis satisfying

$$a_i \circ b_j = -b_j \circ a_i = \delta_{ij}, \qquad a_i \circ a_j = 0, \qquad b_i \circ b_j = 0.$$

Let $\{w_1\, d\lambda, w_2\, d\lambda\}$ be the normalized basis of holomorphic differentials determined by

$$\oint_{a_j} w_i\, d\lambda = \delta_{ij}, \qquad \oint_{b_j} w_i\, d\lambda = B_{ij}, \tag{2.5}$$

for $i, j = 1, 2$.

For a fixed base point $P_0 \in \mathscr{F}_2$, the Abel map is defined by

$$\mathcal{A}_{P_0}(P) = \left(\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P)\right)^{\top} = \left(\int_{P_0}^{P} w_1\, d\lambda,\ \int_{P_0}^{P} w_2\, d\lambda \right)^{\top}. \tag{2.6}$$

Let $d\Omega_1, d\Omega_2, d\Omega_3$ be normalized Abelian differentials satisfying

$$\oint_{a_j} d\Omega_k = 0, \qquad j = 1, 2, \quad k = 1, 2, 3. \tag{2.7}$$

Their corresponding Abelian integrals admit the asymptotic expansions

$$\Omega_1(P) = \pm \left(\ln \lambda + \ln \omega_1 + o(1) \right), \qquad P \to \infty^{\pm}, \tag{2.8}$$

$$\Omega_2(P) = \pm \left(\lambda + \omega_2 + o(1) \right), \qquad P \to \infty^{\pm}, \tag{2.9}$$

and

$$\Omega_3(P) = \pm \left(2\lambda^2 + \omega_3 + o(1) \right), \qquad P \to \infty^{\pm}. \tag{2.10}$$

Fix the base point $P_0 = (\lambda_3, 0)$, and define

$$\Omega_k(P) = \int_{P_0}^{P} d\Omega_k, \qquad k = 1, 2, 3. \tag{2.11}$$

The corresponding b-period vectors are defined by

$$U_k = \frac{1}{2\pi \mathrm{i}} \oint_{b_k} d\Omega_2, \qquad V_k = \frac{1}{2\pi \mathrm{i}} \oint_{b_k} d\Omega_3, \qquad \Delta_k = -\frac{1}{2\pi \mathrm{i}} \oint_{b_k} d\Omega_1, \tag{2.12}$$

for $k = 1, 2$. If ℓ is a positively oriented circle surrounding ∞^+, then

$$\oint_{\ell} d\Omega_1 = 2\pi \mathrm{i}. \tag{2.13}$$

The vectors $U = (U_1, U_2)$ and $V = (V_1, V_2)$ satisfy

$$U_1 = U_2 \in \mathrm{i}\mathbb{R}, \tag{2.14}$$

and

$$V_1 = -V_2 \in \mathbb{R}. \tag{2.15}$$

Moreover,

$$\Delta_k \in \mathbb{R}, \qquad \Delta_1 + \Delta_2 = 1, \tag{2.16}$$

and the symmetry relation

$$\sigma_1 \Delta = -\Delta + \mathbf{1}, \qquad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{2.17}$$

holds.

Define

$$\Omega = D + \mathrm{i}Ux + \mathrm{i}Vt = \mathrm{i}U(x - x_0) + \mathrm{i}V(t - t_0), \tag{2.18}$$

where $D = -\mathrm{i}Ux_0 - \mathrm{i}Vt_0$, and let

$$\Delta = \mathcal{A}_{P_0}(\infty^+) - \mathcal{A}_{P_0}(\infty^-) = \mathcal{A}_{\infty^-}(\infty^+). \tag{2.19}$$

Then

$$\sigma_1 \Omega^* = \Omega, \qquad \mathcal{A}_{\infty^-}(P_0) = \frac{\Delta}{2}, \qquad \mathcal{A}_{\infty^+}(P_0) = -\frac{\Delta}{2}. \tag{2.20}$$

The Baker–Akhiezer function associated with the two-phase solution is given by

(2.21)
$$\begin{pmatrix} \Phi_{11}(P;x,t) \\ \Phi_{21}(P;x,t) \end{pmatrix} = \exp\left[-\mathrm{i}\left(\omega_2 x + \omega_3 t + \frac{\theta}{2} + \frac{\mathrm{i}}{2}\Omega_1(P)\right)\sigma_3\right] \begin{pmatrix} \Theta\big(\Omega + \mathcal{A}_{\infty^+}(P)\big) \\ \Theta\big(\Omega + \Delta + \mathcal{A}_{\infty^+}(P)\big) \end{pmatrix} \frac{\exp\big(\mathrm{i}\Omega_2(P)x + \mathrm{i}\Omega_3(P)t\big)}{\Theta(\Omega)}.$$

Here $\omega_2 = 0$ and $\omega_3 \in \mathbb{R}$.

Using (2.20), we obtain

$$\Omega + \mathcal{A}_{\infty^+}(P) = \Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P), \tag{2.22}$$

and

$$\Omega + \Delta + \mathcal{A}_{\infty^+}(P) = \Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P). \tag{2.23}$$

Define the branch points

$$P_j = (\lambda_j, 0), \qquad P_{-j} = (\lambda_j^*, 0), \qquad j = 1,2,3. \tag{2.24}$$

For $P \in \mathscr{F}_2$, define

$$\boldsymbol{P}_1(P) = \begin{pmatrix} \Phi_{11}(P) \\ \Phi_{21}(P) \end{pmatrix} \left(\Phi_{11}^*\big(\tau_R(\iota(P))\big), \quad \Phi_{21}^*\big(\tau_R(\iota(P))\big)\right), \tag{2.25}$$

and

$$\boldsymbol{S}_1(P) = \begin{pmatrix} \Phi_{21}(P)\,\Phi_{11}^*\big(\tau_R(\iota(P))\big) \\ \Phi_{11}(P)\,\Phi_{21}^*\big(\tau_R(\iota(P))\big) \end{pmatrix}. \tag{2.26}$$

Here τ_R and ι denote the involutions

$$\tau_R : (\lambda, y) \mapsto (\lambda^*, y^*), \qquad \iota : (\lambda, y) \mapsto (\lambda, -y). \tag{2.27}$$

Let

$$\mathscr{P} : \mathscr{F}_2 \to \mathbb{CP}^1, \qquad \mathscr{P}(\lambda, y) = \lambda, \tag{2.28}$$

be the canonical projection map. We consider $\boldsymbol{S}_1(P)$ at the Weierstrass points satisfying

$$\mathscr{P}(P) = \pm\lambda_1, \pm\lambda_2, \pm\lambda_3.$$

Since P is a branch point, one has $\iota(P) = P$. Hence

$$\boldsymbol{S}_1(P) = \begin{pmatrix} \Phi_{21}(P)\,\Phi_{11}^*\big(\tau_R(P)\big) \\ \Phi_{11}(P)\,\Phi_{21}^*\big(\tau_R(P)\big) \end{pmatrix}. \tag{2.29}$$

We now establish the following result concerning the squared eigenfunctions evaluated at the Weierstrass points.

Proposition 1. *The vectors*

$$S_1(P_j), \qquad j = \pm 1, \pm 2, \pm 3,$$

span a three-dimensional subspace. More precisely,

$$\dim\Big(\operatorname{span}\{S_1(P_j) : j = \pm 1, \pm 2, \pm 3\}\Big) = 3. \tag{2.30}$$

Define

$$\begin{pmatrix} \widetilde{\Phi}_{11}(P) \\ \widetilde{\Phi}_{21}(P) \end{pmatrix} = \exp\left[-\mathrm{i}\left(\omega_2 x + \omega_3 t + \frac{\theta}{2} + \frac{\mathrm{i}}{2}\Omega_1(P)\right)\sigma_3\right] \begin{pmatrix} \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P)\Big) \\ \Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P)\Big) \end{pmatrix}. \tag{2.31}$$

Taking complex conjugates at the reflected point $\tau_R(P)$, we obtain

$$\begin{aligned} &\big(\widetilde{\Phi}^*_{11}(\tau_R(P)) \quad \widetilde{\Phi}^*_{21}(\tau_R(P))\big) \\ &= \left(\Theta^*\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P))\right), \quad \Theta^*\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P))\right)\right) \\ &\quad \times \exp\left[\mathrm{i}\left(\omega_2 x + \omega_3 t + \frac{\theta}{2} - \frac{\mathrm{i}}{2}\Omega_1^*(\tau_R(P))\right)\sigma_3\right]. \end{aligned} \tag{2.32}$$

Substituting these expressions into the definition of $S_1(P)$ yields

$$S_1(P) = \widetilde{S}_1(P)\frac{\exp\big(\mathrm{i}(\Omega_2(P) - \Omega_2^*(\tau_R(P)))x + \mathrm{i}(\Omega_3(P) - \Omega_3^*(\tau_R(P)))t\big)}{|\Theta(\Omega)|^2}. \tag{2.33}$$

Here

$$\widetilde{S}_1(P) = \begin{pmatrix} \widetilde{\Phi}_{21}(P)\, \widetilde{\Phi}^*_{11}(\tau_R(P)) \\ \widetilde{\Phi}_{11}(P)\, \widetilde{\Phi}^*_{21}(\tau_R(P)) \end{pmatrix}. \tag{2.34}$$

Using (2.31) and (2.32), we further obtain

$$\begin{aligned} \widetilde{S}_1(P) = \exp&\left[\mathrm{i}\left(2\omega_2 x + 2\omega_3 t + \theta + \frac{\mathrm{i}}{2}\big(\Omega_1(P) - \Omega_1^*(\tau_R(P))\big)\right)\sigma_3\right] \\ &\times \begin{pmatrix} \Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P)\Big)\, \Theta^*\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P))\Big) \\ \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P)\Big)\, \Theta^*\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P))\Big) \end{pmatrix}. \end{aligned} \tag{2.35}$$

Since

$$\mathcal{A}_{P_0}(P) = \begin{pmatrix} \displaystyle\int_{P_0}^{P} w_1\, d\lambda \\ \displaystyle\int_{P_0}^{P} w_2\, d\lambda \end{pmatrix}, \qquad P_0 = P_3 = (\lambda_3, 0), \tag{2.36}$$

we choose all integration paths in the sheet $\mathscr{F}^-$. Then

$$\mathcal{A}_{P_0}(P_3) = 0. \tag{2.37}$$

For the branch point P_2, the integration path corresponds to one half of the cycle b_1. Hence

$$\mathcal{A}_{P_0}(P_2) = \frac{1}{2}\oint_{b_1}\begin{pmatrix} w_1\, d\lambda \\ w_2\, d\lambda \end{pmatrix} = \frac{Be_1}{2}, \tag{2.38}$$

where

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Similarly, the path from P_3 to P_1 is homologous to one half of the cycle $-a_1+b_1$, and therefore

$$\mathcal{A}_{P_0}(P_1) = \frac{1}{2}\oint_{-a_1+b_1}\begin{pmatrix} w_1\,d\lambda \\ w_2\,d\lambda \end{pmatrix} = \frac{B\boldsymbol{e}_1-\boldsymbol{e}_1}{2}. \tag{2.39}$$

For the conjugate branch point P_{-3}, the integration path is homologous to one half of the cycle $-a_1-a_2$. Thus

$$\mathcal{A}_{P_0}(P_{-3}) = \frac{1}{2}\oint_{-a_1-a_2}\begin{pmatrix} w_1\,d\lambda \\ w_2\,d\lambda \end{pmatrix} = -\frac{\boldsymbol{e}_1+\boldsymbol{e}_2}{2} = -\frac{\boldsymbol{1}}{2}, \tag{2.40}$$

where

$$\boldsymbol{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Proceeding in the same way, we obtain

$$\mathcal{A}_{P_0}(P_{-2}) = \frac{1}{2}\oint_{b_2-a_1-a_2}\begin{pmatrix} w_1\,d\lambda \\ w_2\,d\lambda \end{pmatrix} = \frac{B\boldsymbol{e}_2-\boldsymbol{1}}{2}, \tag{2.41}$$

and

$$\mathcal{A}_{P_0}(P_{-1}) = \frac{1}{2}\oint_{b_2-a_1}\begin{pmatrix} w_1\,d\lambda \\ w_2\,d\lambda \end{pmatrix} = \frac{B\boldsymbol{e}_2-\boldsymbol{e}_1}{2}. \tag{2.42}$$

Since $P_0 = P_3$, we immediately obtain

$$\Omega_1(P_3) = \int_{P_0}^{P_3} d\Omega_1 = 0. \tag{2.43}$$

Let ℓ be a positively oriented small circle surrounding ∞^+. Since

$$\oint_{a_j} d\Omega_1 = 0,$$

the integration path from $P_0 = P_3$ to P_{-3} is homologous to one half of the cycle ℓ. Hence

$$\Omega_1(P_{-3}) = \frac{1}{2}\oint_{\ell} d\Omega_1 = \frac{1}{2}\cdot 2\pi\mathrm{i} = \pi\mathrm{i}. \tag{2.44}$$

Similarly, the integration paths corresponding to P_1 and P_2 are homologous to one half of the cycle $\ell+b_1$. Hence

$$\Omega_1(P_1) = \Omega_1(P_2) = \frac{1}{2}\oint_{\ell+b_1} d\Omega_1 = \pi\mathrm{i} + \frac{1}{2}\oint_{b_1} d\Omega_1 = \pi\mathrm{i} - \pi\mathrm{i}\Delta_1 = \pi\mathrm{i}(1-\Delta_1). \tag{2.45}$$

In the same way,

$$\Omega_1(P_{-1}) = \Omega_1(P_{-2}) = \frac{1}{2}\oint_{\ell+b_2} d\Omega_1 = \pi\mathrm{i} + \frac{1}{2}\oint_{b_2} d\Omega_1 = \pi\mathrm{i} - \pi\mathrm{i}\Delta_2 = \pi\mathrm{i}(1-\Delta_2). \tag{2.46}$$

Since

$$\Omega_1(P_j) \in \mathrm{i}\mathbb{R}, \qquad \Delta_1+\Delta_2 = 1,$$

we obtain

$$\Omega_1(P_j) - \Omega_1^*\big(\tau_R(P_j)\big) = \Omega_1(P_j) + \Omega_1(P_{-j}) = \pi\mathrm{i}, \qquad j = \pm1, \pm2, \pm3. \tag{2.47}$$

Consequently,

$$\frac{\mathrm{i}}{2}\Big(\Omega_1(P_j) - \Omega_1^*\big(\tau_R(P_j)\big)\Big) = \frac{\mathrm{i}}{2}\pi\mathrm{i} = -\frac{\pi}{2}. \tag{2.48}$$

By analogous arguments, one also obtains

$$\Omega_2(P_j) - \Omega_2^*\big(\tau_R(P_j)\big) = 0, \qquad \Omega_3(P_j) - \Omega_3^*\big(\tau_R(P_j)\big) = 0, \qquad j = \pm1, \pm2, \pm3. \tag{2.49}$$

Substituting these identities into (2.33), we conclude that

$$\boldsymbol{S}_1(P_j) = \frac{\widetilde{S}_1(P_j)}{|\Theta(\Omega)|^2}, \qquad j = \pm1, \pm2, \pm3. \tag{2.50}$$

Define

$$\widetilde{\theta}_1 = 2\omega_2 x + 2\omega_3 t + \theta - \frac{\pi}{2}. \tag{2.51}$$

Then (2.35) becomes

$$\widetilde{S}_1(P_j) = \exp\big(\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\begin{pmatrix}\Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta^*\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_{-j})\Big)\\ \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta^*\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_{-j})\Big)\end{pmatrix}. \tag{2.52}$$

Using the identities

$$\Theta^*(\boldsymbol{z}) = \Theta(\boldsymbol{z}^*), \qquad \Theta(\boldsymbol{z}) = \Theta(\sigma_1\boldsymbol{z}), \qquad \sigma_1\Omega^* = \Omega,$$

we may rewrite (2.52) as

$$\widetilde{\boldsymbol{S}}_1(P_j) = \exp\big(\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\begin{pmatrix}\Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega - \dfrac{\sigma_1\Delta}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\\ \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega + \dfrac{\sigma_1\Delta}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\end{pmatrix}. \tag{2.53}$$

Since

$$\sigma_1\Delta = -\Delta + \mathbf{1},$$

we further obtain

$$\widetilde{S}_1(P_j) = \exp\big(\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\begin{pmatrix}\Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega + \dfrac{\Delta}{2} - \dfrac{\mathbf{1}}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\\ \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega - \dfrac{\Delta}{2} + \dfrac{\mathbf{1}}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\end{pmatrix}. \tag{2.54}$$

Finally, using the periodicity relation

$$\Theta(\boldsymbol{z} + \boldsymbol{e}_k) = \Theta(\boldsymbol{z}), \qquad k = 1, 2,$$

we conclude from (2.54) that

$$\widetilde{S}_1(P_j) = \exp\big(\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\begin{pmatrix}\Theta\Big(\Omega + \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega + \dfrac{\Delta}{2} + \dfrac{\mathbf{1}}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\\ \Theta\Big(\Omega - \dfrac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\Big)\,\Theta\Big(\Omega - \dfrac{\Delta}{2} + \dfrac{\mathbf{1}}{2} + \sigma_1\mathcal{A}^*_{P_0}(P_{-j})\Big)\end{pmatrix}. \tag{2.55}$$

Note that

$$\mathcal{A}_{P_0}(P_j) + \sigma_1\mathcal{A}^*_{P_0}(P_{-j}) + \frac{\mathbf{1}}{2} = \mathbf{0}, \qquad j = \pm1, \pm2, \pm3. \tag{2.56}$$

For example, using (2.38) and (2.41), we obtain

$$\mathcal{A}_{P_0}(P_2) + \sigma_1\mathcal{A}^*_{P_0}(P_{-2}) + \frac{\mathbf{1}}{2} = \frac{B\boldsymbol{e}_1}{2} + \sigma_1\frac{-B\boldsymbol{e}_2 - \mathbf{1}}{2} + \frac{\mathbf{1}}{2} = \frac{B\boldsymbol{e}_1 - \sigma_1 B\boldsymbol{e}_2}{2} = \mathbf{0}. \tag{2.57}$$

Similarly,

$$\mathcal{A}_{P_0}(P_{-2}) + \sigma_1\mathcal{A}^*_{P_0}(P_2) + \frac{\mathbf{1}}{2} = \frac{B\boldsymbol{e}_2 - \mathbf{1}}{2} + \sigma_1\frac{-B\boldsymbol{e}_1}{2} + \frac{\mathbf{1}}{2} = \frac{B\boldsymbol{e}_2 - \sigma_1 B\boldsymbol{e}_1}{2} = \mathbf{0}. \tag{2.58}$$

Here we have used the symmetry relations

$$B\boldsymbol{e}_1 = \sigma_1 B\boldsymbol{e}_2, \qquad B_{ij} \in \mathrm{i}\mathbb{R}. \tag{2.59}$$

Introduce the notation

$$\Theta^{\boldsymbol{m}}_{\boldsymbol{n}}(\boldsymbol{z}, B) = \begin{pmatrix}\Theta\Big(\boldsymbol{z} + \dfrac{\Delta}{2} + \dfrac{\boldsymbol{n} + B\boldsymbol{m}}{2}, B\Big)\,\Theta\Big(\boldsymbol{z} + \dfrac{\Delta}{2} - \dfrac{\boldsymbol{n} + B\boldsymbol{m}}{2}, B\Big)\\ \Theta\Big(\boldsymbol{z} - \dfrac{\Delta}{2} + \dfrac{\boldsymbol{n} + B\boldsymbol{m}}{2}, B\Big)\,\Theta\Big(\boldsymbol{z} - \dfrac{\Delta}{2} - \dfrac{\boldsymbol{n} + B\boldsymbol{m}}{2}, B\Big)\end{pmatrix}. \tag{2.60}$$

Using (2.55) together with (2.56), we first obtain

$$\widetilde{S}_1(P_3) = \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\begin{pmatrix}\Theta\left(\Omega + \frac{\Delta}{2}\right)^2\\ \Theta\left(\Omega - \frac{\Delta}{2}\right)^2\end{pmatrix} = \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\mathbf{0}}^{\mathbf{0}}(\Omega, B). \tag{2.61}$$

For P_{-3}, using (2.40), we have

$$\begin{aligned}\widetilde{S}_1(P_{-3}) &= \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\begin{pmatrix}\Theta\left(\Omega + \frac{\Delta}{2} - \frac{\mathbf{1}}{2}\right)\Theta\left(\Omega + \frac{\Delta}{2} + \frac{\mathbf{1}}{2}\right)\\ \Theta\left(\Omega - \frac{\Delta}{2} - \frac{\mathbf{1}}{2}\right)\Theta\left(\Omega - \frac{\Delta}{2} + \frac{\mathbf{1}}{2}\right)\end{pmatrix}\\ &= \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\mathbf{1}}^{\mathbf{0}}(\Omega, B).\end{aligned} \tag{2.62}$$

Similarly,

$$\begin{aligned}\widetilde{S}_1(P_2) &= \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\begin{pmatrix}\Theta\left(\Omega + \frac{\Delta}{2} + \frac{B\boldsymbol{e}_1}{2}\right)\Theta\left(\Omega + \frac{\Delta}{2} - \frac{B\boldsymbol{e}_1}{2}\right)\\ \Theta\left(\Omega - \frac{\Delta}{2} + \frac{B\boldsymbol{e}_1}{2}\right)\Theta\left(\Omega - \frac{\Delta}{2} - \frac{B\boldsymbol{e}_1}{2}\right)\end{pmatrix}\\ &= \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\mathbf{0}}^{\boldsymbol{e}_1}(\Omega, B).\end{aligned} \tag{2.63}$$

In the same way,

$$\widetilde{S}_1(P_{-2}) = \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\mathbf{1}}^{\boldsymbol{e}_2}(\Omega, B), \tag{2.64}$$

and

$$\widetilde{S}_1(P_1) = \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\boldsymbol{e}_1}^{\boldsymbol{e}_1}(\Omega, B), \qquad \widetilde{S}_1(P_{-1}) = \exp(\mathrm{i}\widetilde{\theta}_1\sigma_3)\Theta_{\boldsymbol{e}_1}^{\boldsymbol{e}_2}(\Omega, B). \tag{2.65}$$

The Riemann addition formula states that

$$\Theta(\boldsymbol{z} + \boldsymbol{w}, B)\Theta(\boldsymbol{z} - \boldsymbol{w}, B) = \sum_{\varepsilon\in\{0,1\}^2}\Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\boldsymbol{w}, 2B)\Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\boldsymbol{z}, 2B). \tag{2.66}$$

Here the theta function with characteristics is defined by

$$\Theta\begin{bmatrix}\varepsilon\\ \varepsilon'\end{bmatrix}(\boldsymbol{z}, B) = \exp\left(2\pi\mathrm{i}\left(\frac{\varepsilon^\top B\varepsilon}{8} + \frac{\varepsilon^\top\boldsymbol{z}}{2} + \frac{\varepsilon^\top\varepsilon'}{4}\right)\right)\Theta\left(\boldsymbol{z} + \frac{\varepsilon' + B\varepsilon}{2}, B\right), \tag{2.67}$$

and

$$\{0,1\}^2 = \{\mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \mathbf{1}\}. \tag{2.68}$$

Let

$$\boldsymbol{w} = \frac{\boldsymbol{n} + B\boldsymbol{m}}{2}. \tag{2.69}$$

Applying (2.66) to each component of $\Theta_{\boldsymbol{n}}^{\boldsymbol{m}}(\boldsymbol{z}, B)$, we obtain

$$\Theta_{\boldsymbol{n}}^{\boldsymbol{m}}(\boldsymbol{z}, B) = \sum_{\varepsilon\in\{0,1\}^2}\Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(\boldsymbol{n} + B\boldsymbol{m}, 2B)\begin{pmatrix}\Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\boldsymbol{z} + \Delta, 2B)\\ \Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\boldsymbol{z} - \Delta, 2B)\end{pmatrix}. \tag{2.70}$$

Define

$$f_j(x,t) = f_\varepsilon(\Omega) = \begin{pmatrix} \Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\Omega+\Delta, 2B) \\ \Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(2\Omega-\Delta, 2B) \end{pmatrix}, \qquad \varepsilon \in \{\mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \mathbf{1}\}, \tag{2.71}$$

where $j = 1,2,3,4$ corresponds respectively to $\varepsilon = \mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \mathbf{1}$.

Next, define

$$\widetilde{a}_{jk} = \widetilde{a}_{\varepsilon,(\boldsymbol{m},\boldsymbol{n})} = \Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(\boldsymbol{n} + B\boldsymbol{m}, 2B), \tag{2.72}$$

where

$$\varepsilon = \mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \mathbf{1}, \qquad k = 1,2,\ldots,6,$$

and

$$\begin{pmatrix}\boldsymbol{m}\\ \boldsymbol{n}\end{pmatrix} = \begin{pmatrix}\mathbf{0}\\ \mathbf{0}\end{pmatrix}, \quad \begin{pmatrix}\mathbf{0}\\ \mathbf{1}\end{pmatrix}, \quad \begin{pmatrix}\boldsymbol{e}_1\\ \mathbf{0}\end{pmatrix}, \quad \begin{pmatrix}\boldsymbol{e}_1\\ \boldsymbol{e}_1\end{pmatrix}, \quad \begin{pmatrix}\boldsymbol{e}_2\\ \boldsymbol{e}_1\end{pmatrix}, \quad \begin{pmatrix}\boldsymbol{e}_2\\ \mathbf{1}\end{pmatrix}. \tag{2.73}$$

For $k = 1,\ldots,6$, define

$$g_k(x,t) = \exp\big(-\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\widetilde{\boldsymbol{S}}_1(P_j), \qquad j = 3,-3,2,1,-1,-2, \tag{2.74}$$

respectively. Equivalently,

$$g_1 = \exp\big(-\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\widetilde{\boldsymbol{S}}_1(P_3), \quad g_2 = \exp\big(-\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\widetilde{\boldsymbol{S}}_1(P_{-3}), \quad \ldots, \quad g_6 = \exp\big(-\mathrm{i}\widetilde{\theta}_1\sigma_3\big)\widetilde{\boldsymbol{S}}_1(P_{-2}).$$

Then (2.55) and (2.70) imply that

$$g_k = \sum_{j=1}^{4} \widetilde{a}_{jk} f_j, \qquad k = 1,2,\ldots,6. \tag{2.75}$$

Equivalently,

$$[g_1, g_2, \ldots, g_6] = [f_1, f_2, f_3, f_4]\widetilde{A}, \qquad \widetilde{A}_{jk} = \widetilde{a}_{jk}. \tag{2.76}$$

Since the functions f_j are linearly independent and $\exp\big(\mathrm{i}\widetilde{\theta}_1\sigma_3\big)$ is an invertible diagonal matrix, multiplying by this factor does not affect linear dependence. Hence it suffices to determine $\operatorname{rank}(\widetilde{A})$.

Using (2.67), we compute

$$\begin{aligned} \Theta\begin{bmatrix}\varepsilon\\ \mathbf{0}\end{bmatrix}(\boldsymbol{n} + B\boldsymbol{m}, 2B) &= \exp\left(2\pi\mathrm{i}\left(\frac{\varepsilon^\top B\varepsilon}{4} + \frac{\varepsilon^\top(\boldsymbol{n}+B\boldsymbol{m})}{2}\right)\right)\Theta(\boldsymbol{n} + B\boldsymbol{m} + B\varepsilon, 2B) \\ &= \exp\left(\frac{\pi\mathrm{i}}{2}\varepsilon^\top B\varepsilon\right)\exp\Big(\pi\mathrm{i}\,\varepsilon^\top(\boldsymbol{n}+B\boldsymbol{m})\Big)\Theta\big(B(\boldsymbol{m}+\varepsilon), 2B\big). \end{aligned} \tag{2.77}$$

Motivated by this expression, define

$$a_{jk} = a_{\varepsilon,(\boldsymbol{m},\boldsymbol{n})} = \exp\Big(\pi\mathrm{i}\,\varepsilon^\top(\boldsymbol{n}+B\boldsymbol{m})\Big)\Theta\big(B(\boldsymbol{m}+\varepsilon), 2B\big), \tag{2.78}$$

for $j = 1,\ldots,4$ and $k = 1,\ldots,6$. Then

$$\widetilde{A} = \operatorname{diag}\left(1, \exp\left(\frac{\pi\mathrm{i}}{2}\boldsymbol{e}_1^\top B\boldsymbol{e}_1\right), \exp\left(\frac{\pi\mathrm{i}}{2}\boldsymbol{e}_2^\top B\boldsymbol{e}_2\right), \exp\left(\frac{\pi\mathrm{i}}{2}\mathbf{1}^\top B\mathbf{1}\right)\right) A, \tag{2.79}$$

where $A_{jk} = a_{jk}$.

Denote

$$\Theta_{ij} = \Theta\big(B(i\boldsymbol{e}_1 + j\boldsymbol{e}_2), 2B\big), \qquad i,j \in \{0,1\}. \tag{2.80}$$

From (2.78), we first obtain

$$a_{j1} = \Theta(B\varepsilon, 2B), \qquad a_{j2} = \exp\Big(\pi\mathrm{i}\varepsilon^\top\mathbf{1}\Big)\Theta(B\varepsilon, 2B). \tag{2.81}$$

Similarly,

$$a_{j3} = \exp\Big(\pi \mathrm{i} \varepsilon^\top B \boldsymbol{e}_1\Big)\, \Theta(B(\boldsymbol{e}_1+\varepsilon), 2B)\,, \tag{2.82}$$

and

$$a_{j4} = \exp\Big(\pi \mathrm{i} \varepsilon^\top (\boldsymbol{e}_1 + B\boldsymbol{e}_1)\Big)\, \Theta(B(\boldsymbol{e}_1+\varepsilon), 2B)\,. \tag{2.83}$$

Moreover,

$$a_{j5} = \exp\Big(\pi \mathrm{i} \varepsilon^\top (\boldsymbol{e}_1 + B\boldsymbol{e}_2)\Big)\, \Theta(B(\boldsymbol{e}_2+\varepsilon), 2B)\,, \tag{2.84}$$

and

$$a_{j6} = \exp\Big(\pi \mathrm{i} \varepsilon^\top (\mathbf{1} + B\boldsymbol{e}_2)\Big)\, \Theta(B(\boldsymbol{e}_2+\varepsilon), 2B)\,. \tag{2.85}$$

Using the ordering

$$\varepsilon = \mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \mathbf{1},$$

the first two column vectors of A are

$$A_1 = \begin{pmatrix} \Theta_{00} \\ \Theta_{10} \\ \Theta_{01} \\ \Theta_{11} \end{pmatrix}, \qquad A_2 = \begin{pmatrix} \Theta_{00} \\ -\Theta_{10} \\ -\Theta_{01} \\ \Theta_{11} \end{pmatrix}. \tag{2.86}$$

For the third column, we compute

$$A_3 = \begin{pmatrix} \Theta_{10} \\ e^{\pi \mathrm{i} B_{11}} e^{-2\pi \mathrm{i} B_{11}} \Theta_{00} \\ e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ e^{\pi \mathrm{i}(B_{11}+B_{21})} e^{-2\pi \mathrm{i}(B_{11}+B_{21})} \Theta_{01} \end{pmatrix} = \begin{pmatrix} \Theta_{10} \\ e^{-\pi \mathrm{i} B_{11}} \Theta_{00} \\ e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ e^{-\pi \mathrm{i}(B_{11}+B_{21})} \Theta_{01} \end{pmatrix}. \tag{2.87}$$

Similarly,

$$A_4 = \begin{pmatrix} \Theta_{10} \\ -\, e^{-\pi \mathrm{i} B_{11}} \Theta_{00} \\ e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ -\, e^{-\pi \mathrm{i}(B_{11}+B_{21})} \Theta_{01} \end{pmatrix}. \tag{2.88}$$

Proceeding in the same way, we obtain

$$A_5 = \begin{pmatrix} \Theta_{01} \\ -\, e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ e^{\pi \mathrm{i} B_{22}} e^{-2\pi \mathrm{i} B_{22}} \Theta_{00} \\ -\, e^{\pi \mathrm{i}(B_{12}+B_{22})} e^{-2\pi \mathrm{i}(B_{12}+B_{22})} \Theta_{10} \end{pmatrix} = \begin{pmatrix} \Theta_{01} \\ -\, e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ e^{-\pi \mathrm{i} B_{11}} \Theta_{00} \\ -\, e^{-\pi \mathrm{i}(B_{21}+B_{11})} \Theta_{10} \end{pmatrix}. \tag{2.89}$$

Likewise,

$$A_6 = \begin{pmatrix} \Theta_{01} \\ -\, e^{\pi \mathrm{i} B_{21}} \Theta_{11} \\ -\, e^{-\pi \mathrm{i} B_{11}} \Theta_{00} \\ e^{-\pi \mathrm{i}(B_{21}+B_{11})} \Theta_{10} \end{pmatrix}. \tag{2.90}$$

Hence the matrix A takes the form

$$A = [A_1, A_2, \ldots, A_6]. \tag{2.91}$$

Applying the column transformations

$$A \longrightarrow \left[\frac{A_1+A_2}{2}, \frac{A_1-A_2}{2}, \frac{A_3+A_4}{2}, \frac{A_3-A_4}{2}, \frac{A_5+A_6}{2}, \frac{A_5-A_6}{2}\right],$$

the matrix A is transformed into

$$\begin{pmatrix} \Theta_{00} & 0 & \Theta_{10} & 0 & \Theta_{01} & 0 \\ 0 & \Theta_{10} & 0 & e^{-\pi \mathrm{i} B_{11}}\Theta_{00} & -e^{\pi \mathrm{i} B_{21}}\Theta_{11} & 0 \\ 0 & \Theta_{01} & e^{\pi \mathrm{i} B_{21}}\Theta_{11} & 0 & 0 & e^{-\pi \mathrm{i} B_{11}}\Theta_{00} \\ \Theta_{11} & 0 & 0 & e^{-\pi \mathrm{i}(B_{11}+B_{21})}\Theta_{01} & 0 & -e^{-\pi \mathrm{i}(B_{21}+B_{11})}\Theta_{10} \end{pmatrix}. \tag{2.92}$$

Introduce the notation

$$a = \Theta_{00}, \qquad b = \Theta_{10}, \qquad c = \Theta_{01}, \qquad d = \Theta_{11}. \tag{2.93}$$

Then (2.92) becomes

$$\begin{pmatrix} a & 0 & b & 0 & c & 0 \\ 0 & b & 0 & e^{-\pi \mathrm{i} B_{11}}a & -e^{\pi \mathrm{i} B_{21}}d & 0 \\ 0 & c & e^{\pi \mathrm{i} B_{21}}d & 0 & 0 & e^{-\pi \mathrm{i} B_{11}}a \\ d & 0 & 0 & e^{-\pi \mathrm{i}(B_{11}+B_{21})}c & 0 & -e^{-\pi \mathrm{i}(B_{21}+B_{11})}b \end{pmatrix}. \tag{2.94}$$

We now perform elementary row operations. First, eliminate the entry d in the fourth row and first column by replacing

$$R_4 \longrightarrow R_4 - \frac{d}{a}R_1.$$

Next, eliminate the entry c in the third row and second column by replacing

$$R_3 \longrightarrow R_3 - \frac{c}{b}R_2.$$

Finally, eliminate the remaining nonzero entries in the fourth row using the second and third rows. This yields

$$\begin{pmatrix} a & 0 & b & 0 & c & 0 \\ 0 & b & 0 & e^{-\pi \mathrm{i} B_{11}}a & -e^{\pi \mathrm{i} B_{21}}d & 0 \\ 0 & 0 & e^{\pi \mathrm{i} B_{21}}d & -e^{-\pi \mathrm{i} B_{11}}\frac{ac}{b} & e^{\pi \mathrm{i} B_{21}}\frac{cd}{b} & e^{-\pi \mathrm{i} B_{11}}a \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.95}$$

Therefore, $\mathrm{rank}(A) = 3$. Since $\widetilde{A}$ differs from A only by multiplication with an invertible diagonal matrix, one also has $\mathrm{rank}(\widetilde{A}) = 3$. Therefore Proposition 1 follows:

$$\dim\Big(\mathrm{span}\{S_1(P_j) : j = \pm 1, \pm 2, \pm 3\}\Big) = \dim\Big(\mathrm{span}\{g_k : k = 1, 2, \ldots, 6\}\Big) = 3. \tag{2.96}$$

HW 3: Numerical Reproduction of One-Mode Anomalous-Wave Recurrence **2026-05-20**

In this homework we reproduce the first numerical figure in the paper of P. G. Grinevich and P. M. Santini, *The finite-gap method and the periodic NLS Cauchy problem of anomalous waves for a finite number of unstable modes*, Russian Mathematical Surveys **74** (2019), 211–263. The main purpose is not only to draw a similar picture, but also to explain how the numerical simulation is built from the equation. I write the explanation in a beginner-oriented way: first the continuous model, then the discrete grid, then the time stepping formula, and finally the MATLAB implementation.

The equation considered in the paper is the focusing nonlinear Schrödinger equation

$$\mathrm{i}u_t + u_{xx} + 2|u|^2 u = 0, \qquad u = u(x,t) \in \mathbb{C}, \tag{3.1}$$

with periodic boundary conditions in x. Here x is the spatial variable, t is time, and the unknown $u(x,t)$ is complex-valued. The quantity that we plot is the amplitude $|u(x,t)|$. The exact constant-amplitude background is

$$u_0(x,t) = \mathrm{e}^{2\mathrm{i}t}, \tag{3.2}$$

so $|u_0(x,t)| = 1$ for all x and t. The numerical experiment starts from a very small periodic perturbation of the unit background at $t=0$, and the interesting question is whether this tiny perturbation stays tiny or grows into a large visible wave.

Modulation instability. Modulation instability is the growth of a small long-wave perturbation of a plane wave background. For the focusing NLS equation (3.1), write the solution near the background in the form

$$u(x,t) = \mathrm{e}^{2\mathrm{i}t}\left(1 + w(x,t)\right), \qquad |w| \ll 1. \tag{3.3}$$

After linearization, each Fourier pair with wave number k evolves as a combination of a growing and a decaying mode when

$$0 < k < 2. \tag{3.4}$$

More precisely, the growth rate is

$$\sigma(k) = k\sqrt{4-k^2}, \qquad 0 < k < 2. \tag{3.5}$$

If $k > 2$, the corresponding linearized frequency is real and the mode does not grow exponentially. This is the first important idea behind the picture: not every Fourier mode is dangerous. Only the modes whose wave numbers lie in the interval $0 < k < 2$ are linearly unstable. Thus the focusing sign in (3.1) is essential: it is the reason why a perturbation of size 10^{-4} can later become an $O(1)$ coherent wave.

On a periodic interval of length L, the admissible Fourier wave numbers are

$$k_j = \frac{2\pi j}{L}, \qquad j = 1, 2, \ldots. \tag{3.6}$$

Therefore $k_j < 2$ is equivalent to $j < L/\pi$, and the number of unstable positive Fourier modes is

$$N = \left\lfloor \frac{L}{\pi} \right\rfloor. \tag{3.7}$$

This is the basic mechanism behind the phrase "a finite number of unstable modes" in the paper.

One unstable mode. The initial condition used for Figure 1 of the paper is

$$u(x,0) = 1 + \epsilon\left(c_1 \mathrm{e}^{\mathrm{i}k_1 x} + c_{-1}\mathrm{e}^{-\mathrm{i}k_1 x}\right), \qquad k_1 = \frac{2\pi}{L}, \tag{3.8}$$

where

$$L = 6, \qquad \epsilon = 10^{-4}, \qquad c_1 = \frac{1}{2}, \qquad c_{-1} = \frac{0.3 - 0.4\mathrm{i}}{2}. \tag{3.9}$$

For these parameters,

$$N = \left\lfloor \frac{L}{\pi} \right\rfloor = 1, \qquad k_1 = \frac{\pi}{3} < 2, \tag{3.10}$$

so only the first Fourier mode belongs to the unstable band (3.4). The second mode would have wave number $2k_1 = 2\pi/3 > 2$, so it is outside the unstable band. Thus this is a clean one-mode test case. The perturbation first grows from the almost constant background, then forms a localized Akhmediev-type peak, and finally returns close to the background before the next recurrence.

Split-step Fourier discretization. The split-step Fourier method is a standard method for equations such as (3.1), because the equation contains two parts that are easy in two different representations. First rewrite the equation as

$$u_t = \mathrm{i}u_{xx} + 2\mathrm{i}|u|^2 u. \tag{3.11}$$

The term $\mathrm{i}u_{xx}$ is the linear dispersive part. It is easy after applying the Fourier transform. The term $2\mathrm{i}|u|^2u$ is the nonlinear part. It is easy at each fixed grid point in physical space. The split-step idea is to solve these two parts one after the other during each small time interval.

The spatial interval in the code is

$$[-L/2, L/2] = [-3, 3],$$

and the code uses $N_x = 384$ grid points. The grid spacing is therefore

$$\Delta x = \frac{L}{N_x}.$$

The grid is periodic, so the left endpoint and the right endpoint represent the same point. This is why the MATLAB code creates $N_x + 1$ points by `linspace` and then removes the last one. The Fourier wave numbers used by MATLAB's FFT are ordered as

$$k = \frac{2\pi}{L}\left(0, 1, \ldots, \frac{N_x}{2} - 1, -\frac{N_x}{2}, \ldots, -1\right). \tag{3.12}$$

This ordering is not arbitrary: it is the ordering expected by the vectors produced by `fft`.

For the linear subproblem

$$u_t = \mathrm{i}u_{xx},$$

write

$$u(x,t) = \sum_k \widehat{u}_k(t)\mathrm{e}^{\mathrm{i}kx}.$$

Since differentiating twice gives

$$u_{xx} = \sum_k (-k^2)\widehat{u}_k(t)\mathrm{e}^{\mathrm{i}kx},$$

each Fourier coefficient satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\widehat{u}_k = -\mathrm{i}k^2\widehat{u}_k.$$

Therefore the exact linear evolution over a time interval τ is

$$\widehat{u}_k(t+\tau) = \mathrm{e}^{-\mathrm{i}k^2\tau}\widehat{u}_k(t). \tag{3.13}$$

In the MATLAB code, one half linear step has $\tau = \Delta t/2$, so the multiplier is

```
Ehalf = exp(-1i*k.^2*dt/2).
```

For the nonlinear subproblem

$$u_t = 2\mathrm{i}|u|^2u,$$

there is no spatial derivative. Thus each grid value evolves independently. During this nonlinear substep the modulus $|u|$ is constant, and only the complex phase rotates. The exact nonlinear evolution over one full time step Δt is

$$u(t+\Delta t) = u(t)\exp\left(2\mathrm{i}|u(t)|^2\Delta t\right). \tag{3.14}$$

This is exactly the MATLAB command

```
u = u .* exp(2i*abs(u).^2*dt).
```

The code uses $T_{\max} = 60$, $N_t = 30000$, and hence

$$\Delta t = \frac{T_{\max}}{N_t} = \frac{60}{30000} = 0.002. \tag{3.15}$$

The number 0.002 is small enough for a smooth reproduction while keeping the runtime short. One full Strang step is

$$u^{n+1} = \mathcal{L}_{\Delta t/2}\mathcal{N}_{\Delta t}\mathcal{L}_{\Delta t/2}u^n, \tag{3.16}$$

where

$$\mathcal{L}_{\Delta t/2}u = \mathcal{F}^{-1}\left[\mathrm{e}^{-\mathrm{i}k^2\Delta t/2}\mathcal{F}u\right], \qquad \mathcal{N}_{\Delta t}u = u\exp(2\mathrm{i}|u|^2\Delta t).$$

Written out in the same order as the MATLAB loop, this is

$$u^{(1)} = \mathcal{F}^{-1}\left[\mathrm{e}^{-\mathrm{i}k^2\Delta t/2}\mathcal{F}u^n\right], \tag{3.17}$$

$$u^{(2)} = u^{(1)}\exp\left(2\mathrm{i}|u^{(1)}|^2\Delta t\right), \tag{3.18}$$

$$u^{n+1} = \mathcal{F}^{-1}\left[\mathrm{e}^{-\mathrm{i}k^2\Delta t/2}\mathcal{F}u^{(2)}\right]. \tag{3.19}$$

Equations (3.17)–(3.19) are the mathematical version of the three main operations inside the `for` loop: FFT half-step, nonlinear phase rotation, FFT half-step.

MATLAB code and explanation. The final script used for the paper-style black-and-white reproduction is `reproduce_paper_fig1_bw_v6.m`. It also saves and reuses `fig1_data.mat`, so the expensive time integration does not need to be repeated every time we only want to adjust the appearance of the figure.

```
function reproduce_paper_fig1_bw_v6()

    clear; clc; close all;

    %% ================================================================
    % Parameters from the paper
    %% ================================================================

    L    = 6;
    eps0 = 1e-4;

    c1   = 1/2;
    cm1  = (0.3 - 0.4i)/2;

    k1 = 2*pi/L;

    %% ================================================================
    % Numerical parameters
    %% ================================================================

    Nx   = 384;
    Tmax = 60;

    Nt = 30000;
    dt = Tmax/Nt;

    saveEvery = 25;

    thisDir = fileparts(mfilename('fullpath'));
    datafile = fullfile(thisDir,'fig1_data.mat');
```

This first block defines the function, clears the MATLAB workspace, and records the parameters from the paper. The period is $L = 6$, the perturbation size is $\epsilon = 10^{-4}$, and the two Fourier coefficients are $c_1 = 1/2$ and $c_{-1} = (0.3 - 0.4\mathrm{i})/2$. The code then sets $k_1 = 2\pi/L$, chooses $N_x = 384$, $T_{\max} = 60$, and $N_t = 30000$. The line `dt = Tmax/Nt` is the time step $\Delta t = 0.002$. The variable `thisDir` is used so that both the cached data file and the exported PNG are saved in the same folder as the MATLAB file.

```
    if exist(datafile,'file')

        load(datafile,'x','Tplot','Uplot','L','Tmax');

    else

        x = linspace(-L/2, L/2, Nx+1);
        x(end) = [];

        k = (2*pi/L)*[0:Nx/2-1, -Nx/2:-1];

        u = 1 + eps0*( ...
              c1  * exp( 1i*k1*x ) ...
            + cm1 * exp(-1i*k1*x ) );

        Ehalf = exp(-1i*k.^2*dt/2);

        nSave = floor(Nt/saveEvery) + 1;

```

```
20        Uplot = zeros(nSave,Nx);
21        Tplot = zeros(nSave,1);
22
23        Uplot(1,:) = abs(u);
24        Tplot(1)   = 0;
```

This block either loads previously computed data or starts a new simulation. If `fig1_data.mat` already exists, MATLAB loads `x`, `Tplot`, and `Uplot`; otherwise it constructs the periodic spatial grid and the FFT wave-number vector (3.12). The initial condition in line 47 is the discrete version of (3.8). Notice that the small factor ϵ appears as `eps0`, so the coefficients `c1` and `cm1` are the paper coefficients themselves, not already scaled coefficients. The array `Uplot` will store $|u|$, while `Tplot` stores the corresponding output times.

```
1         idx = 1;
2
3         for n = 1:Nt
4
5             % half linear step
6             uhat = fft(u);
7             uhat = Ehalf .* uhat;
8             u = ifft(uhat);
9
10            % nonlinear step
11            u = u .* exp(2i*abs(u).^2*dt);
12
13            % half linear step
14            uhat = fft(u);
15            uhat = Ehalf .* uhat;
16            u = ifft(uhat);
17
18            if mod(n,saveEvery)==0
19                idx = idx + 1;
20                Uplot(idx,:) = abs(u);
21                Tplot(idx)   = n*dt;
22            end
23        end
24
25        save(datafile,'x','Tplot','Uplot','L','Tmax');
26
27    end
```

This is the numerical time stepping block. Lines 65–68 perform the first linear half-step: `fft(u)` computes the Fourier coefficients, multiplication by `Ehalf` applies the factor $\exp(-ik^2\Delta t/2)$, and `ifft` returns to physical space. Line 71 applies the nonlinear phase rotation (3.14). Lines 73–76 repeat the same Fourier half-step and complete one Strang step. The code stores the amplitude $|u|$ every `saveEvery = 25` time steps, so the plotted data are not too large but still show a smooth surface.

```
1     [Xgrid,Tgrid] = meshgrid(x,Tplot);
2
3     % Use software OpenGL in batch mode when available.
4     try
5         opengl software;
6     catch
7     end
8
9     % Changed back to v3-style size.
10    figure('Color','w','Position',[80 80 1400 460]);
11    set(gcf,'Renderer','opengl');
12    set(gcf,'InvertHardcopy','off');
13
14    %% ================================================================
15    % Grayscale surface coloring
16    %% ================================================================
17
18    grayLo = 0.44;
19    grayHi = 0.72;
20
21    Umin = 0.70;
22    Umax = 2.85;
23
24    Uclip = min(max(Uplot,Umin),Umax);
25    Uscaled = (Uclip - Umin)/(Umax - Umin);
26
27    Cgray = grayLo + (grayHi-grayLo)*Uscaled;
```

```
s = surf(Tgrid, Xgrid, Uplot, Cgray);

set(s, ...
    'EdgeColor','none', ...
    'FaceColor','interp', ...
    'FaceLighting','gouraud', ...
    'AmbientStrength',0.55, ...
    'DiffuseStrength',0.85, ...
    'SpecularStrength',0.18, ...
    'SpecularExponent',25);

colormap(gray(256));
caxis([0 1]);
```

This block converts the saved numerical data into a surface plot. The horizontal axes are t and x, and the vertical axis is $|u(x,t)|$. The array Cgray is not new numerical data; it is only color information for the surface. It maps the amplitude into a moderate grayscale range so that the final image resembles the black-and-white figure in the paper instead of MATLAB's default colored surface plot.

```
xlim([-4 Tmax+4]);
ylim([-L/2 L/2+0.55]);
zlim([0 3.05]);

% Changed back to v3-style aspect ratio.
pbaspect([9.5 1.55 1.05]);

% Changed back to v3-style orientation.
view([33 18]);
camproj('orthographic');

% Use the v5 framing but remove the excess canvas after export.
camzoom(1.45);

axis off;
box off;

% Make the axes occupy almost the whole PNG.
set(gca,'Position',[-0.03 -0.03 1.08 1.08]);
set(gca,'LooseInset',[0 0 0 0]);

%% ==============================================================
% Lighting
%% ==============================================================

delete(findall(gca,'Type','light'));

light('Position',[-35 -25 35],'Style','infinite');
light('Position',[45 20 18],'Style','infinite');

lighting gouraud;
material dull;
```

This block controls the camera, aspect ratio, zoom, and lighting. These commands do not change the computed solution; they only change how the three-dimensional surface is viewed. The limits include the whole computed amplitude range, the aspect ratio makes the time direction long and thin as in the paper, and the orthographic camera avoids perspective distortion. The two light sources create the gray highlights and shadows that make the peaks easier to see.

```
hold on;

z0 = 0.07;
y0 = -L/2;

plot3([0 Tmax],[y0 y0],[z0 z0], ...
    'k','LineWidth',1.15);

quiver3(0,y0,z0,-3.2,0,0,0, ...
    'k','LineWidth',1.15,'MaxHeadSize',0.8);

quiver3(Tmax,y0,z0,0,L,0,0, ...
    'k','LineWidth',1.15,'MaxHeadSize',0.8);

quiver3(Tmax,y0,z0,0,0,1.15,0, ...
    'k','LineWidth',1.15,'MaxHeadSize',0.8);
```

```
text(-3.7,y0,z0,'t','FontSize',16);
text(Tmax,L/2+0.35,z0,'x','FontSize',16);

hold off;

%% ==============================================================
% Save and crop PNG
%% ==============================================================

drawnow;

set(gcf,'PaperPositionMode','auto');
rawFile = fullfile(thisDir,'paper_fig1_reproduction_bw_v6_raw.png');
outFile = fullfile(thisDir,'paper_fig1_reproduction_bw_v6.png');

print(gcf,rawFile,'-dpng','-r300');

rgb = imread(rawFile);
contentMask = any(rgb < 248,3);

rows = find(any(contentMask,2));
cols = find(any(contentMask,1));

if ~isempty(rows) && ~isempty(cols)
    pad = 80;
    r1 = max(min(rows)-pad,1);
    r2 = min(max(rows)+pad,size(rgb,1));
    c1 = max(min(cols)-pad,1);
    c2 = min(max(cols)+pad,size(rgb,2));
    rgb = rgb(r1:r2,c1:c2,:);
end

imwrite(rgb,outFile);

if exist(rawFile,'file')
    delete(rawFile);
end
```

The last block draws simple black coordinate arrows and exports the PNG file. The arrows are added with `plot3`, `quiver3`, and `text`; they are visual annotations, not part of the numerical method. The v6 script first writes a temporary raw PNG, reads it back with `imread`, detects the non-white part of the image, and crops away the large empty white margins. It then writes the final v6 PNG. This changes only the exported canvas, not the computed solution.

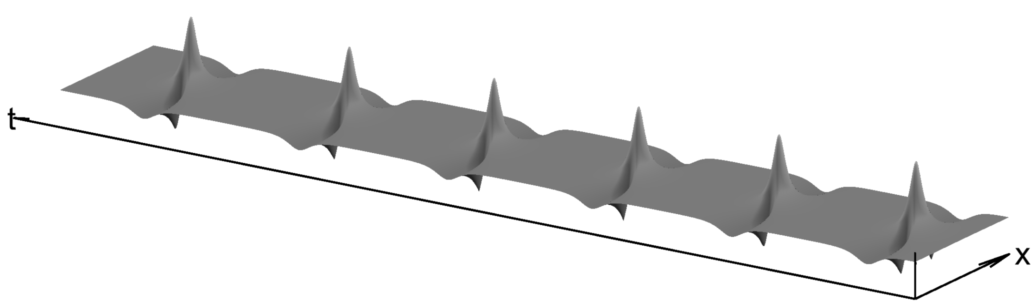


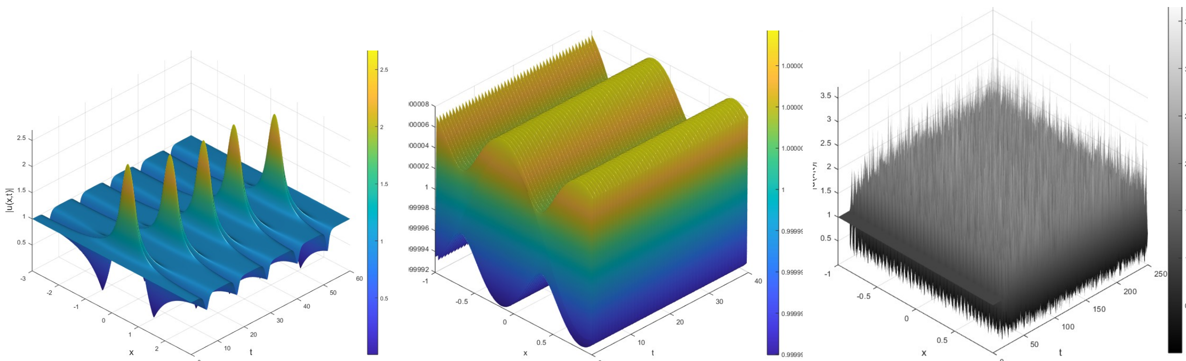
FIGURE 1. Paper-style black-and-white reproduction produced by `reproduce_paper_fig1_bw_v6.m`. This version uses the same split-step Fourier data as before, but crops the MATLAB export so that the three-dimensional recurrence fills the canvas more effectively while keeping the same LaTeX figure width.

Figure 1 shows the expected anomalous-wave recurrence. For small t, the surface is close to the background $|u| = 1$. The one unstable Fourier mode then grows, a sharp nonlinear peak forms, and the solution returns close to the background before the next growth stage. The important point is that the large peak is not inserted by hand: it is generated by repeatedly applying the three Strang-splitting operations described above.

Supplementary views and stability checks. The earlier colored view is kept because it gives a complementary way to read the recurrence times: the nearly uniform background corresponds to $|u| \approx 1$, while the localized high-amplitude regions correspond to the nonlinear peaks in the paper-style surface. To check that the large peaks are not merely plotting artifacts, we also ran the same type of split-step Fourier simulation in two stable regimes. In the first stable test, the code keeps a fine grid and small time step, but changes the length to $L = 2$. Then

$$k_1 = \frac{2\pi}{L} = \pi > 2. \tag{3.20}$$

Thus the first Fourier mode is outside the modulationally unstable band. With a small perturbation and conservative numerical parameters, the solution remains close to the unit background. The under-resolved test uses a coarser grid $N_x = 256$, a much larger final time $T_{\max} = 250$, and only $N_t = 20000$ time steps; its irregular spikes should therefore be interpreted as numerical drift or accumulated discretization error, not as physical modulation instability. The three checks are collected compactly in Figure 2.



(a) Colored recurrence view (b) Stable well-resolved run (c) Under-resolved artifact run

FIGURE 2. Supplementary views and stability checks. Panel (a) gives a colored view of the one-mode recurrence, panel (b) shows a stable well-resolved run, and panel (c) shows that coarse long-time simulations can create spurious peaks even in a stable regime.

This compact comparison separates the genuine one-mode recurrence in Figure 1 from artifacts caused by insufficient numerical resolution.

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